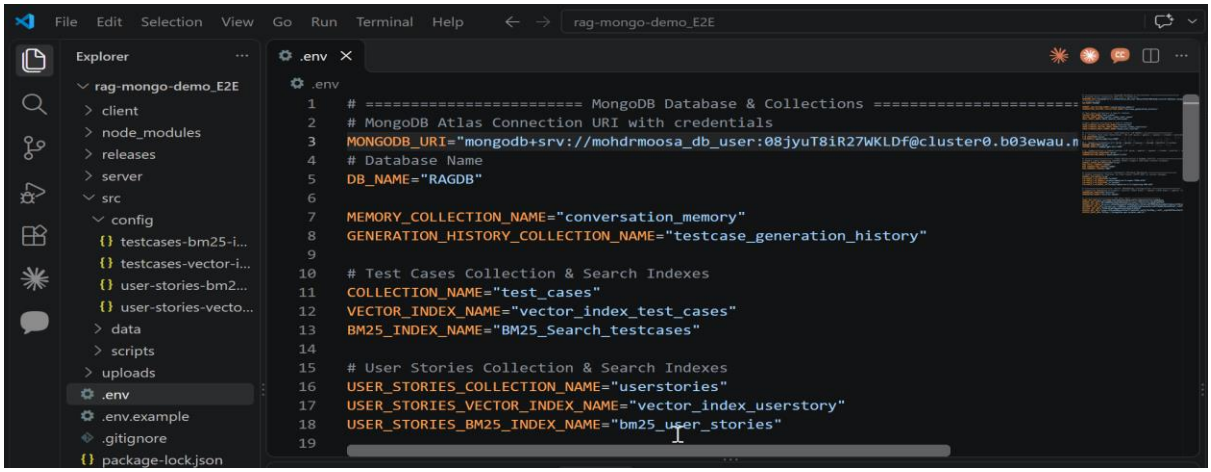
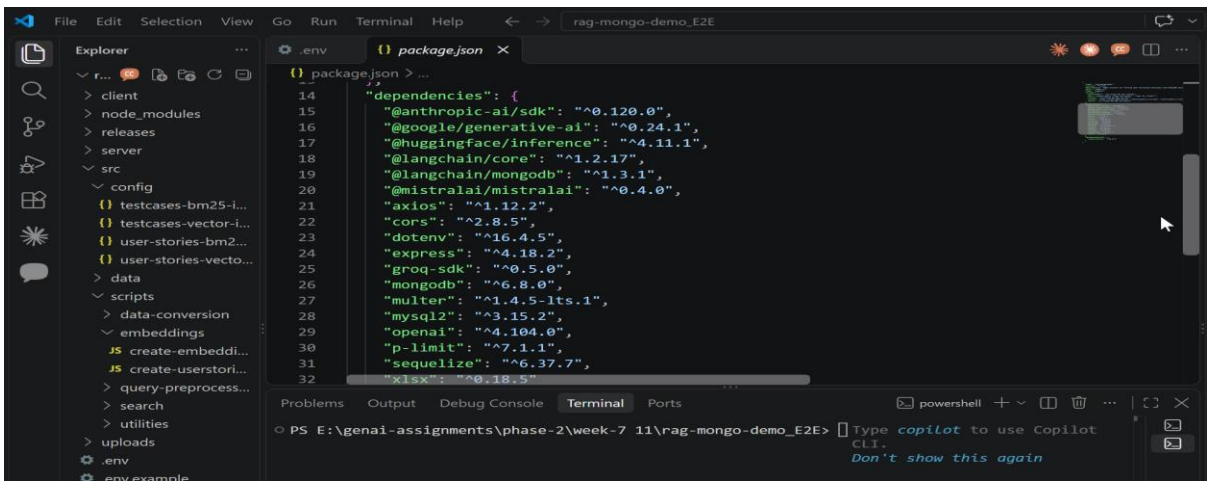
LangChain Persistant Memory – Please Scroll down to see the enhanced RAG Based Test Case Generator –application that maintains persistent conversation memory and stores test case generation history in a dedicated MongoDB collection.

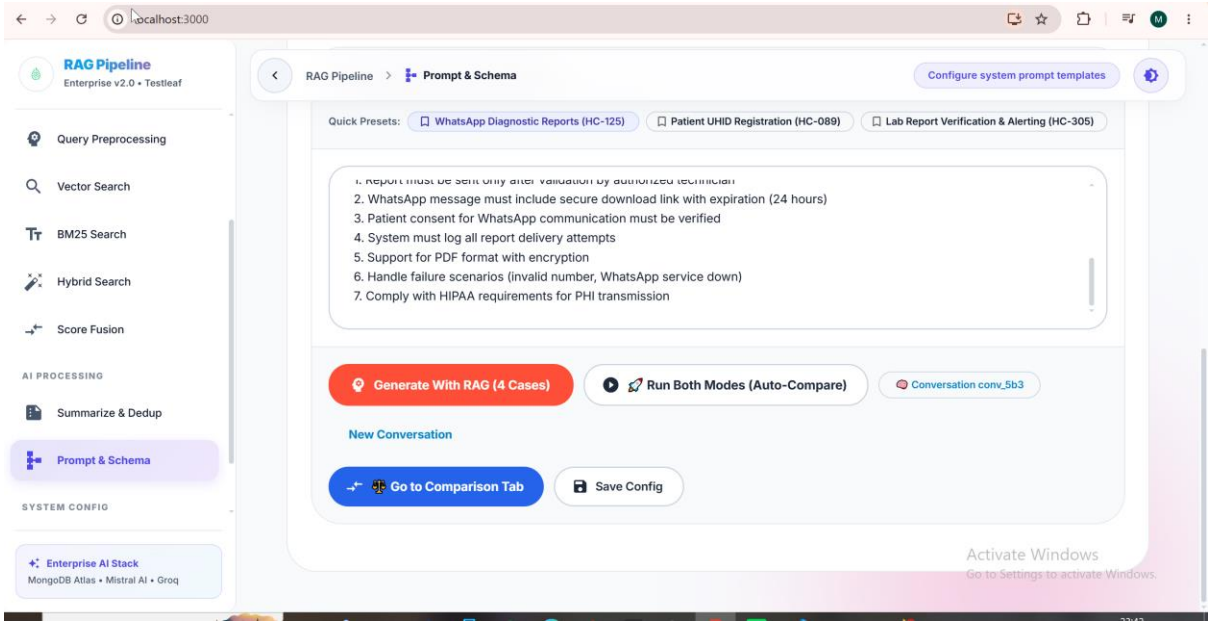
Requirement	How it's met
1. LangChain integration	A custom LangChain chat model sends calls through your existing multi-provider and fallback code, used with a `ChatPromptTemplate`
2. Separate MongoDB collection	`conversation_memory` and `testcase_generation_history` in `RAGDB`, indexed, set up when the server starts
3. Persistent memory	`MongoDBChatMessageHistory` stores compact memory per conversation and mode; the last 5 exchanges are replayed; the conversation ID survives browser restarts
4. Unique identifiers	`conv_`, `exec_` and `mem_` IDs, shown in the UI and stored in memory and history
5. Generation metadata	Query, test cases, conversation and execution details, date and time, prompt and raw response

.env file updated for conversation_memory and testcase_generation_history

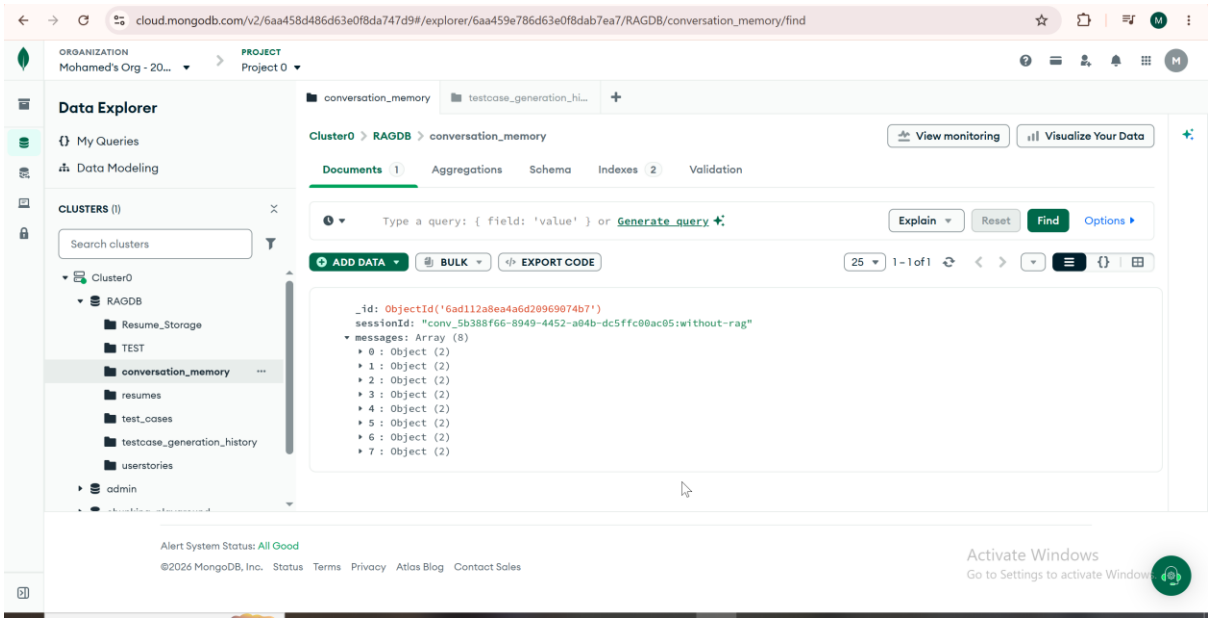


Langchain LLM Model version

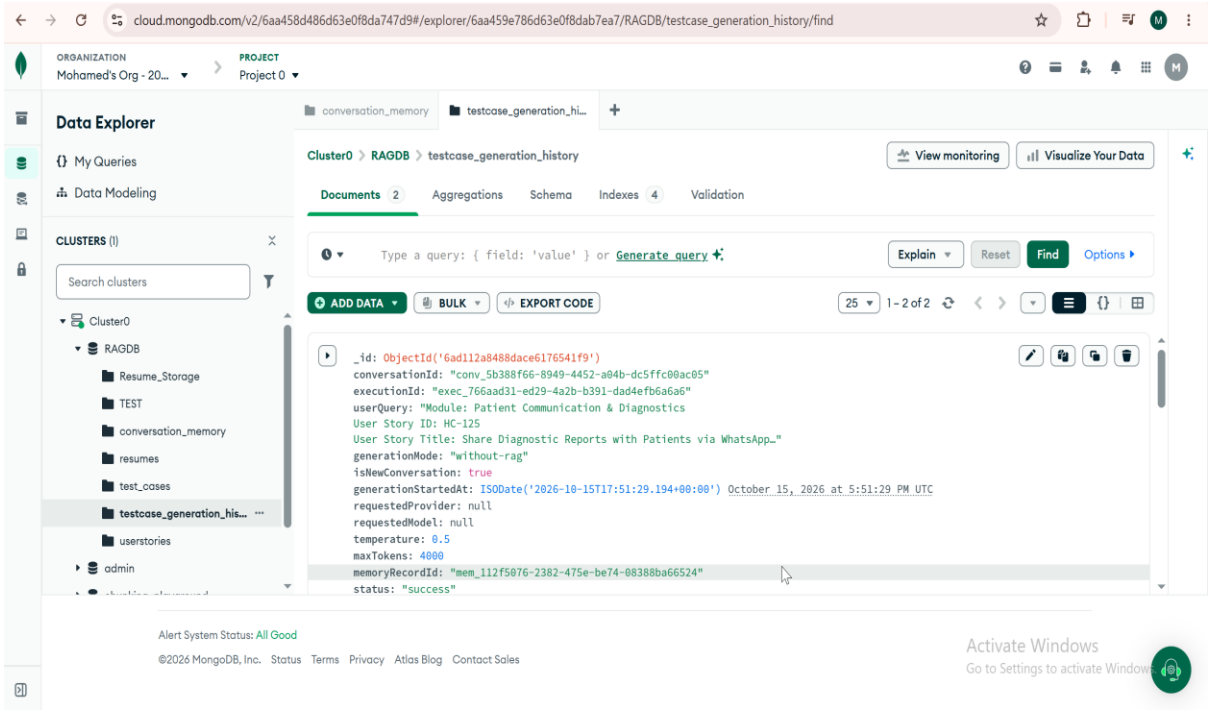




Conversation memory TC generation without -rag



Testcase generation history – along with Conversation ID stored in MongoDB



Expected outcome

An enhanced RAG Test Case Generator application that maintains persistent conversation memory and stores test case generation history in a dedicated MongoDB collection.

Part of the outcome	Evidence from your own runs
Enhanced RAG Test Case Generator	Generation runs through LangChain; your existing UI, providers and fallback keep working
Maintains persistent conversation memory	`conversation_memory` holds 8 messages across 4 runs; each run replayed the earlier ones; the conversation ID is kept across browser restarts
Stores test case generation history	4 records containing user query, test cases, conversation and execution details, date and time, prompt and raw response
In a dedicated MongoDB collection	`testcase_generation_history` (and `conversation_memory`), separate from `test_cases` and `user_stories`

Requirements checklist

#	Requirement	Status
1	LangChain integration	✓

#	Requirement	Status
2	Separate MongoDB collection	✓
3	Persistent memory across sessions	✓
4	Unique IDs (`conv_`, `exec_`, `mem_`)	✓
5	Generation metadata	✓

Testcase generation history – Conversation ID stored in MongoDB

